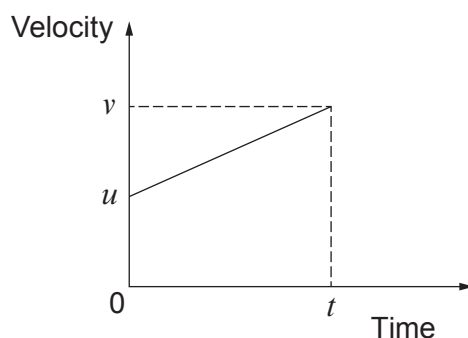


6. (a) A velocity-time graph is given for a body which is accelerating uniformly for a time, t .



Using the graph, show clearly that:

[3]

$$x = ut + \frac{1}{2}at^2$$

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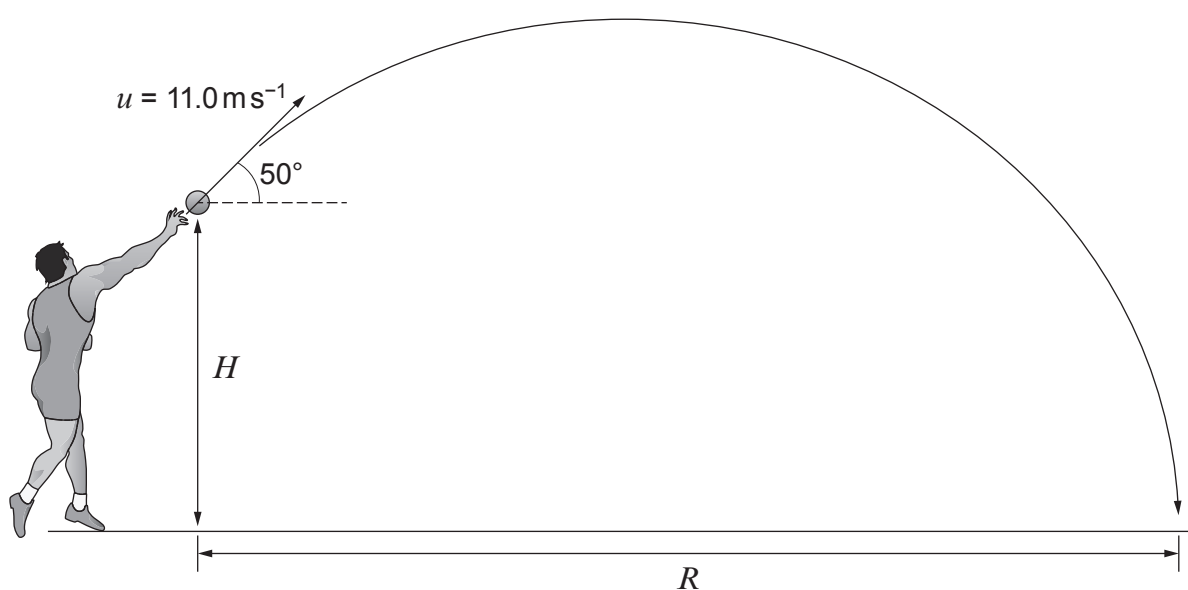
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- (b) The shot put is an event in athletics. The shot (a heavy metal ball) is thrown from a standing position so that it lands on the ground a horizontal distance, R , from the thrower.



Steve is a shot putter. He throws a shot with a velocity of 11.0 m s^{-1} in a direction 50° to the horizontal. The shot takes 1.95 s to reach the ground.

- (i) Show that the vertical component of the initial velocity is approximately 8 m s^{-1} . [1]

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- (ii) Hence calculate, H , the vertical distance above the ground from which the shot was thrown. [3]

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- (iii) Calculate the horizontal distance, R , for this throw. [2]

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- (c) Steve makes the following comment to Bryn, a fellow competitor.

“Throwing the shot at an angle even greater than 50° to the horizontal will increase the horizontal distance travelled by the shot because the time of flight will increase.”

Consider whether or not it would be a good idea for Bryn to follow Steve’s advice. Justify your answer. [3]

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